Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Whitmore, Mihriban Ph.D.		
Project Title:	Development and Demonstration of an Integrated, Independent Human Factors Evaluation Process for Space Payloads and Equipment		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mihriban.whitmore-1@nasa.gov	**Fax:**	FY 281-244-5773
PI Organization Type:	NASA CENTER	**Phone:**	281-244-1004
Organization Name:	NASA - Johnson Space Center		
PI Address 1:	Mail Code SF3		
PI Address 2:	2101 NASA Road 1		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	10/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Norris, Lena (Lockheed Martin)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	At NASA, and in everyday life experience, there are multiple examples of equipment that has passed through a rigorous requirements validation process, but which nevertheless is shown to have major usability difficulties upon implementation and actual use. At NASA, these usability difficulties can have a major impact on very precious resources—crew time on-orbit and the time of ground controllers. In extreme cases, such as responding to emergencies, usability difficulties can even threaten the physical well being of the crew. The project proposed here covers one way to improve the ultimate usability of fielded NASA equipment and systems, and thus enhance the productivity of crew and ground controllers—namely the use of an integrated, life-cycle human factors evaluation process. An integrated human factors evaluation process will be proposed that focuses on the unique NASA context and addresses NASA equipment and payloads, along with their associated procedures, user's manuals,		

Task Description:	displays, labels, and training. The draft Human Factors process document has just been completed. The benefits of this process will then be demonstrated via pilot demonstrations (evaluations) with a NASA equipment or payload in the early design stage. The hardware will be selected such that physical and cognitive interfaces can be evaluated, along with related procedures, owner's manuals and labeling. Human factors professionals, crew, and payload engineers and procedures writers will participate in the demonstration. Data from the pilot demonstration will be analyzed to determine the beneficial impacts from using the human factors process. Lessons learned from the pilot demonstration will be incorporated in the final version of the process document.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Development of a practical Human Factors Engineering (HFE) process and demonstration of its effectiveness will help other HFE personnel integrate HFE processes in their own work environment.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)